



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

HOLE DEVOUR IO

CSA0920-PROGRAMMING IN JAVA

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Review 1

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GAME CONCEPT OVERVIEW

What is Hole Devour IO?

- A 2D arcade game where the player controls a black hole.
- Consume smaller objects to grow in size.
- Each consumed object increases the score.
- Core Gameplay: Move → Consume → Grow → Score → Win.
- Win by becoming the largest hole.

REFERENCE GAME STUDY



Reference Game – Hole.io

- Popular arcade .io game.
- Consume objects to grow.
- Mouse-based movement.
- Size determines what can be consumed.

Our Simplified Version

- Single-player mode.
- Basic collision detection.
- Fixed game map.
- Simple object spawning.



OBJECTIVES / LEARNING OUTCOMES



- Understand Object-Oriented Programming.
- Implement Java Collections Framework.
- Design a Swing-based game interface.
- Implement collision detection.
- Store game results using JDBC.

PROPOSED MODULES



Module 1:

Map & Object Spawner

- Generate random objects on the map.
- Store all objects in the game.
- Respawn objects after consumption.

Java Concept

- Collections Framework (ArrayList)

PROPOSED MODULES

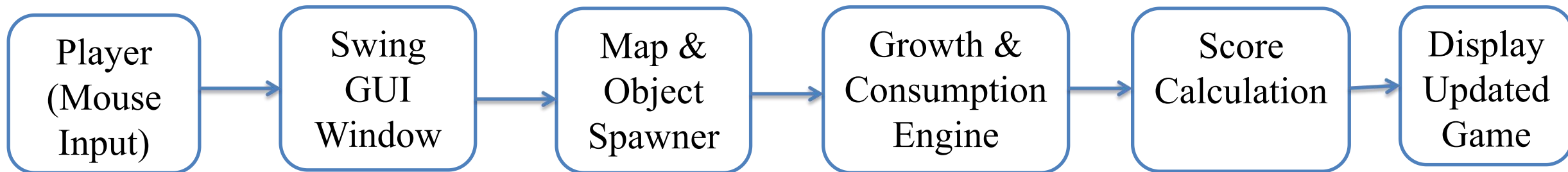
Module 2

- Hole Growth & Consumption Engine
- Detect collisions with objects.
- Consume smaller objects.
- Increase hole size.
- Update player score.

Java Concept

- Object-Oriented Programming (Classes & Objects)

SYSTEM ARCHITECTURE



TECH STACK & WHY

Technology	Purpose
Java	Game development
OOP	Hole & Object Classes
Collections	Store game objects
Swing	User Interface
JDBC	Save progress (future module)

PROPOSED DATABASE SCHEMA

Table: MATCH_RESULT

Column	Data Type
Player_ID	INT
Player_Name	VARCHAR(50)
Score	INT
Hole_Size	DOUBLE
Match_Date	DATE

Stored Data

- Player information
- Final score
- Hole size



MULTITHREADING PLAN



- Keep the game responsive.
- Handle future AI movement.
- Support smooth gameplay without freezing.

EXCEPTION HANDLING PLAN

Situation	Exception	Solution
Invalid collision	IllegalArgumentException	Ignore collision
Missing object	NullPointerException	Respawn object
Invalid game action	Exception	Display error message



WORK PLAN



Week 1

- Design game concept.
- Create object classes.

Week 2

- Implement Object Spawner.
- Test object generation.

Week 3

- Develop Hole Growth Engine.
- Build Swing GUI.

Week 4

- JDBC integration.
- Testing and debugging.

ANTICIPATED CHALLENGES

Challenge 1

- Implement accurate collision detection.

Solution: Use distance-based collision logic.

Challenge 2

- Manage game objects efficiently.

Solution: Use ArrayList for dynamic storage.

Challenge 3

- Maintain smooth gameplay.

Solution: Optimize Swing rendering and event handling.

Thank You!